NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2.

SATURDAY 29TH JUNE, 2019 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are two sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(d) ***Indicate your names, birth date, gender, class, school and district on all your answer sheets****.*

(e) *Answer scripts of only qualified students should be returned to the contest coordinator, latest Monday 1st July, 2019.*

(f) *ALL participants who have sat for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 27*th July,* 2019 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each

Qn. 1. Suppose x, y and z are three real numbers such that $3x$, $4y$, $5z$ is a geometric sequence and $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ is an arithmetic sequence. What is the value of $\frac{x}{z} + \frac{z}{x}$?

Qn. 2. An equilateral triangle is inscribed in a circle of area $1m^2$. Then the second circle is inscribed in the triangle. Find the radius of the second circle.

Qn. 3. Given $x^2 + xy + y^2 = 84$ and $x - \sqrt{xy} + y = 6$, determine the value of xy.

Qn. 4. Four cards are laid out in front of you. You know for sure that on one side of each card is a single number, and on the other side of each card is a single geometric shape. The same number or the same geometric shape might be found on more than one of these four cards.

You see (on the top side) respectively a 2, a 5, a triangle, and a square. Your friend says: "Every card with a square has a 4 on the other side." Your task is to determine whether your friend is correct by choosing some of the cards to be flipped over. The chosen cards will only be flipped after you have made your choice(s). What is the fewest number of cards you can choose if you want to know for sure whether or not your friend is correct?

Qn. 5. List all real values of x which satisfy

$$\frac{6}{\sqrt{x-8}-9} + \frac{1}{\sqrt{x-8}-4} + \frac{7}{\sqrt{x-8}+4} + \frac{12}{\sqrt{x-8}+9} = 0$$

Qn. 6. Let $F_0 = 0$ and $F_1 = 1$. For all $n \geq 2$, define F_n to be the remainder of $F_{n-1} + F_{n-2}$ divided by 3. The sequence starts as follows: $0, 1, 1, 2, 0, 2, \ldots$ What is

$$F_{2017} + F_{2018} + F_{2019} + F_{2020} + F_{2021} + F_{2022} + F_{2023} + F_{2024}?$$

Qn. 7. Let ABC be a triangle with $\angle BAC \neq 90^0$. Let O be the circumcenter of the triangle ABC and let Γ be the circumcircle of the triangle BOC: Suppose that Γ intersects the line segment AB at P different from B; and the line segment AC at Q different from C: Let ON be a diameter of the circle Γ: Prove that the quadrilateral $APNQ$ is a parallelogram.

Qn. 8. How many integer triples (x, y, z) satisfy the following equation $x^2 + y^2 + z^2 = 2xyz$.

Qn. 9. How many different ways can one order the integers 1 through 5 so that no three consecutive integers in the ordering are in increasing order?

Qn. 10. Let $f(x) = \dfrac{x}{\sqrt{1+x^2}}$ and let $f_n(x) = \underbrace{f(f(f(\cdots(f}_{n}(x))\cdots)))$. In other words, $f_1(x) = f(x)$ and then we recursively define $f_{n+1}(x)$ as $f(f_n(x))$. What is $f_{99}(1)$?

SECTION B : 10 marks each

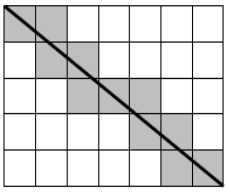
Qn. 11. Given a sequence $\{a_n\}$ of numbers satisfying $a_0 = 1$, $a_{n+1} = \dfrac{7a_n + \sqrt{45a_n^2 - 36}}{2}$, $n \in \mathbb{N}$. Show that

(a) for each $n \in \mathbb{N}$, a_n is a positive integer.

(b) for each $n \in \mathbb{N}$, $a_n \cdot a_{n+1} - 1$ is a perfect square.

Qn. 12. Let ABC be an acute triangle with $\angle BAC = 30^0$. The internal and external angle bisectors of $\angle ABC$ meet the line AC at B_1 and B_2 respectively, and the internal and external angle bisectors of $\angle ACB$ meet the line AB at C_1 and C_2 respectively. Suppose that the circles with diameters B_1B_2 and C_1C_2 meet inside the triangle ABC at point P.
Prove that $\angle BPC = 90^0$.

Qn. 13. A diagonal of this 5×7 rectangle passes through 11 squares. (We say that a line passes through a square if the square and the line have at least two common points.) How many square will the diagonal of a 2016×2018 rectangle pass through?



Qn. 14. (a) Suppose f is a function such that for every real number x, $f(x) + f(1-x) = 10$ and $f(1+x) = 4 + f(x)$. What is the value of $f(100) + f(-100)$?

(b) Find all positive real numbers x and y that satisfy the following system of equations:

$$\begin{aligned} x^y &= y^{x-y} \\ x^x &= y^{12y} \end{aligned}$$

Qn. 15. You and your partner went to a dinner party in which there were four other couples. After the dinner was over, you asked everyone except yourself: "How many people did you shake hands with tonight?" To your surprise, no two people gave the same number, so that someone did not shake any hands, someone else shook only one person's hand, a third person only shook two people's hands, and so on. Assume nobody shook hands with their own partner or with themselves. How many people did your partner shake hands with that evening at the party?

NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2 SOLUTIONS.

SATURDAY 29TH JUNE, 2019 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) *This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.*

(b) *There are two sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*).*

(c) *Marks will be awarded for only answers for which a clear and logical layout of the working is provided. A correct solution, poorly presented, will not earn full marks.*

(d) ***Indicate your names, birth date, gender, class, school and district on all your answer sheets**.*

(e) *Answer scripts of only qualified students should be returned to the contest coordinator, latest Monday 1st July, 2019.*

(f) *ALL participants who have sat for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 27*th July,* 2019 *at 8:00am at Makerere University, Kampala.*

SECTION A : 5 marks each

Qn. 1. Suppose x, y and z are three real numbers such that $3x$, $4y$, $5z$ is a geometric sequence and $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ is an arithmetic sequence. What is the value of $\frac{x}{z} + \frac{z}{x}$?

Solution : *Since* $3x, 4y, 5z$ *is a geometric sequence,* $\frac{4y}{3x} = \frac{5z}{4y}$*, so that*

$$16y^2 = 15xz \tag{1.1}$$

Also, since $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ *is an arithmetic sequence,*

$$\frac{1}{y} - \frac{1}{x} = \frac{1}{z} - \frac{1}{y}$$

or

$$\frac{2}{y} = \frac{1}{x} + \frac{1}{z} = \frac{x+z}{xz}.$$

Which imply that

$$x + z = \frac{2xz}{y} \tag{1.2}$$

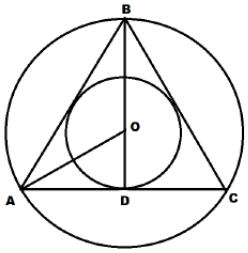
Therefore,

$$\begin{aligned}
\frac{x}{z}+\frac{z}{x} &= \frac{x^2+z^2}{xz} \\
&= \frac{(x+z)^2}{xz}-2 \\
&= \frac{(2xz/y)^2}{xz}-2 \quad \textit{using Eqn (1.2)} \\
&= \frac{4xz}{y^2}-2 \\
&= \frac{64}{14}-2 \quad \textit{using Eqn (1.1)} \\
&= \frac{34}{15}
\end{aligned}$$

■

Qn. 2. An equilateral triangle is inscribed in a circle of area $1m^2$. Then the second circle is inscribed in the triangle. Find the radius of the second circle.

Solution : *In an equilateral triangle the centres of inscribed and circumscribed circles are located at the same point O. AO and BO are bisectors and AD is the height. Therefore AOD is a right triangle and $\angle OAD$ is 30^0. Thus,*



$$|OD| = \frac{|AO|}{2}$$

. Since the area of the circumscribed circle is $1m^2$,

$$\pi|AO|^2 = 1$$

and

$$|AO| = \frac{1}{\sqrt{\pi}}.$$

Therefore, $|OD| = \frac{1}{2\sqrt{\pi}}$. ***Answer: The radius of the second cicle is*** $\frac{1}{2\sqrt{\pi}}m$.

■

Qn. 3. Given $x^2 + xy + y^2 = 84$ and $x - \sqrt{xy} + y = 6$, determine the value of xy.

Solution : *Rewrite the second equation as* $x + y = 6 + \sqrt{xy}$. *Squaring both then gives* $x^2 + y^2 + 2xy = 36 + xy + 12\sqrt{xy}$. *Subtracting this equation from* $x^2 + xy + y^2 = 84$ *and simplifying gives* $48 = 12\sqrt{xy}$, *thus* $xy = 16$ ■

Qn. 4. Four cards are laid out in front of you. You know for sure that on one side of each card is a single number, and on the other side of each card is a single geometric shape. The same number or the same geometric shape might be found on more than one of these four cards.

You see (on the top side) respectively a 2, a 5, a triangle, and a square. Your friend says: "Every card with a square has a 4 on the other side." Your task is to determine whether your friend is correct by choosing some of the cards to be flipped over. The chosen cards will only be flipped after you have made your choice(s). What is the fewest number of cards you can choose if you want to know for sure whether or not your friend is correct?

Solution : *Clearly you must flip over the square, to see if there is a* 4 *on the other side. You must also flip over the* 2 *and the* 5*: if either has a square on the other side, it contradicts your friend's statement. There is no need to flip over the triangle: the other side will have some number, and it doesn't matter whether it's a* 4 *or not. Therefore, the answer is a* 3. ■

Qn. 5. List all real values of x which satisfy

$$\frac{6}{\sqrt{x-8}-9}+\frac{1}{\sqrt{x-8}-4}+\frac{7}{\sqrt{x-8}+4}+\frac{12}{\sqrt{x-8}+9}=0$$

Solution : *Let* $y=\sqrt{x-8}$. *The equation becomes* $\frac{18y-54}{y^2-81}+\frac{8y-24}{y^2-16}=0$, *or* $(y-3)\left(\frac{18}{y^2-81}+\frac{8}{y^2-16}\right)=0$. *Thus* $y=3$ *or* $26y^2-(8\cdot 81+16\cdot 18)=0$. *The second equation yields* $y^2=36$, *so* $y=6$. *Finally* $x=3^2+8$ *or* 6^2+8. *Therefore the values are* 17 *and* 44 ■

Qn. 6. Let $F_0 = 0$ and $F_1 = 1$. For all $n \geq 2$, define F_n to be the remainder of $F_{n-1} + F_{n-2}$ divided by 3. The sequence starts as follows:

$$0, 1, 1, 2, 0, 2, \ldots$$

What is $F_{2017} + F_{2018} + F_{2019} + F_{2020} + F_{2021} + F_{2022} + F_{2023} + F_{2024}$?

Solution : *The first few terms of the sequence are* $0,\ 1,\ 1,\ 2,\ 0,\ 2,\ 2,\ 1,\ 0,\ 1,\ 1,$ *so the sequence is periodic with a period* 8. *Thus* $F_{2017} = F_1$ *since* $2017 = 8.252+1$, $F_{2018} = F_2$, *etc. The sum is* $1 + 1 + 2 + 0 + 2 + 2 + 1 + 0 = 9$. ■

Qn. 7. Let ABC be a triangle with $\angle BAC \neq 90^0$. Let O be the circumcenter of the triangle ABC and let Γ be the circumcircle of the triangle BOC: Suppose that Γ intersects the line segment AB at P different from B; and the line segment AC at Q different from C: Let ON be a diameter of the circle Γ: Prove that the quadrilateral $APNQ$ is a parallelogram.

Solution : *From the assumption that the circle Γ intersects both of the line segments AB and AC, it follows that the 4 points N, C, Q, O are located on Γ in the order of N, C, Q, O or in the order of N, C, O, Q.*
The following argument for the proof of the assertion of the problem is valid in either case. Since $\angle NQC$ and $\angle NOC$ are subtended by the same arc $\widehat{NC}$ of Γ at the points Q and O, respectively, on Γ , we have $\angle NQC = \angle NOC$: We also have $\angle BOC = 2\angle BAC$, since $\angle BOC$ and $\angle BAC$ are subtended by the same arc $\widehat{BC}$ of the circum-circle of the triangle ABC at the center O of the circle and at the point A on the circle, respectively. From $OB = OC$ and the fact that ON is a diameter of Γ, it follows that the triangles OBN and OCN are congruent, and therefore we obtain $2\angle NOC = \angle BOC$:
Consequently, we have $\angle NQC = \frac{1}{2}\angle BOC = \angle BAC$; which shows that the 2 lines AP, QN are parallel. In the same manner, we can show that the 2 lines AQ, PN are also parallel. Thus, the quadrilateral $APNQ$ is a parallelogram. ■

Qn. 8. How many integer triples (x, y, z) satisfy the following equation?

$$x^2 + y^2 + z^2 = 2xyz$$

Solution : *Since the left-hand side is even, either all of x, y, and z, or exactly one of x, y and z are even. If only one term is even, then the right side is divisible by 4 and the left side leaves a remainder of 2 after dividing by 4. So this can't be. Therefore, assume that $x = 2x_1$, $y = 2y_1$, $z = 2z_1$ for integers x, y, and z. Plugging these equations in, we obtain $4x_1^2 + 4y_1^2 + 4z_1^2 = 16x_1y_1z_1$, or $x_1^2 + y_1^2 + z_1^2 = 4x_1y_1z_1$. This is nearly the same as our original equation! By an argument identical to the one above, all of x_1, y_1 and z_1 must again be even.*
The pattern keeps repeating, and so we conclude that x, y, and z are all evenly divisible by an arbitrary power of 2. Therefore they are all zero. ■

Qn. 9. How many different ways can one order the integers 1 through 5 so that no three consecutive integers in the ordering are in increasing order?

Solution : *Let A_1 be the set of all permutations $(a_1, a_2, a_3, a_4, a_5)$ of the integers $1, 2, 3, 4, 5$ such that $a_1 < a_2 < a_3$; let A_2 be the set of all permutations $(a_1, a_2, a_3, a_4, a_5)$ of $1, 2, 3, 4, 5$, such that $a_2 < a_3 < a_4$; and let A_3 be a set of all permutations $(a_1, a_2, a_3, a_4, a_5)$ of the integers $1, 2, 3, 4, 5$ such that $a_3 < a_4 < a_5$. We need to count how many permutations of $1, 2, 3, 4, 5$ are not in the union of the sets A_1, A_2,and A_3.*

First, there are $5! = 120$ permutations of $1, 2, 3, 4, 5$.

Next by inclusion-exclusion

$$| A_1 \cup A_2 \cup A_3 | = | A_1 | + | A_2 | + | A_3 | - | A_1 \cap A_2 | - | A_1 \cap A_3 | - | A_2 \cap A_3 | + | A_1 \cap A_2 \cap A_3 | .$$

We have $|A_1| = |A_2| = |A_3| = \binom{5}{3} \cdot 2 = 20$ (for example, to compute $|A_1|$ note that there are $\binom{5}{3}$ ways to choose a_1, a_2, and a_3 and two ways to pick a_4 out of the remaining two numbers).

Also $A_1 \cap A_2$ is the set of all permutations of $1, 2, 3, 4, 5$ where $a_1 < a_2 < a_3 < a_4$, there are 5 such permutations (there are 5 choices for a_5). Similarly, $A_2 \cap A_3$ is the set of all permutations of $1, 2, 3, 4, 5$ where $a_2 < a_3 < a_4 < a_5$, there are 5 such permutations.

Finally, $A_1 \cap A_2 \cap A_3$ is the set of all permutations of $1, 2, 3, 4, 5$ where $a_1 < a_2 < a_3 < a_4 < a_5$, that is the set $\{1, 2, 3, 4, 5\}$.
Thus, $| A_1 \cup A_2 \cup A_3 | = 20 + 20 + 20 - 5 - 1 - 5 + 1 = 50$.

There are also other alternative methods. ■

Qn. 10. Let $f(x) = \dfrac{x}{\sqrt{1+x^2}}$ and let $f_n(x) = \underbrace{f(f(f(\cdots(f}_{n}(x))\cdots)))$. In other words, $f_1(x) = f(x)$ and then we recursively define $f_{n+1}(x)$ as $f(f_n(x))$. What is $f_{99}(1)$?

Solution : *By recursion formula,*

$$\begin{aligned} f_1(x) = f(x) &= \frac{x}{\sqrt{1+x^2}} \\ f_2(x) = f(f(x)) &= \frac{x}{\sqrt{1+2x^2}} \\ \vdots &= \vdots \\ f_{99}(x) &= \frac{x}{\sqrt{1+99x^2}} \end{aligned}$$

Thus $f_{99}(1) = \dfrac{1}{10}$. ■

SECTION B : 10 marks each

Qn. 11. Given a sequence $\{a_n\}$ of numbers satisfying $a_0 = 1,\ a_{n+1} = \dfrac{7a_n + \sqrt{45a_n^2 - 36}}{2}$, $n \in \mathbb{N}$. Show that

(a) for each $n \in \mathbb{N}$, a_n is a positive integer.

Solution : *By assumption;* $a_1 = 5$ *and* $\{a_n\}$ *is strictly increasing with*

$$2a_{n+1} - 7a_n = \sqrt{45a_n^2 - 36}$$

Square both sides;

$$\begin{aligned}
(2a_{n+1} - 7a_n)^2 &= 45a_n^2 - 36 \\
4a_{n+1}^2 - 28a_n a_{n+1} + 49a_n^2 &= 45a_n^2 - 36 \\
4a_{n+1}^2 - 28a_n a_{n+1} + 4a_n^2 + 36 &= 0 \\
a_{n+1}^2 - 7a_n a_{n+1} + a_n^2 + 9 &= 0 \qquad (1.3) \\
a_n^2 - 7a_{n-1} a_n + a_{n-1}^2 + 9 &= 0 \qquad (1.4)
\end{aligned}$$

From (1.3)-(1.4) we have

$$\begin{aligned}
a_{n+1}^2 - a_n^2 - 7a_n a_{n+1} + 7a_n a_{n+1} + a_n^2 - a_{n-1}^2 &= 0 \\
a_{n+1}^2 - 7a_n a_{n+1} + 7a_n a_{n+1} - a_{n-1}^2 &= 0 \\
a_{n+1}^2 - a_{n-1}^2 &= 7a_n a_{n+1} - 7a_n a_{n+1} \\
(a_{n+1} + a_{n-1})(a_{n+1} - a_{n-1}) &= 7a_n (a_{n+1} - a_{n+1}) \\
(a_{n+1} + a_{n-1}) &= 7a_n
\end{aligned}$$

such that

$$a_{n+1} = 7a_n - a_{n-1} \qquad (1.5)$$

From $a_0 = 1, a_2 = 5$ *and Eqn (1.5), then* a_n *is always a positive integer.* ■

(b) for each $n \in \mathbb{N}$, $a_n \cdot a_{n+1} - 1$ is a perfect square.

Solution : *From Eqn (1.3),*

$$(a_{n+1} + a_n)^2 = 9(a_{n+1} \cdot a_n - 1) \quad \Rightarrow \quad a_{n+1} \cdot a_n - 1 = \left(\frac{a_{n+1} + a_n}{3}\right)^2$$

Therefore, $a_n \cdot a_{n+1} - 1$ *is a perfect square.* ■

Qn. 12. Let ABC be an acute triangle with $\angle BAC = 30^0$. The internal and external angle bisectors of $\angle ABC$ meet the line AC at B_1 and B_2 respectively, and the internal and external angle bisectors of $\angle ACB$ meet the line AB at C_1 and C_2 respectively. Suppose that the circles with diameters B_1B_2 and C_1C_2 meet inside the triangle ABC at point P.
Prove that $\angle BPC = 90^0$.

Solution : *Since $\angle B_1BB_2 = 90^0$, the circle having B_1B_2 as its diameter goes through the points $B : B_1 : B_2$: From $B_1A : B_1C = B_2A : B_2C = BA : BC$, it follows that this circle is the Apolonius circle with the ratio of the distances from the points A and C being $BA : BC$. Since the point P lies on this circle, we have*

$$PA : PC = BA : BC = \sin C : \sin A$$

; from which it follows that $PA \sin A = PC \sin C$: Similarly, we have $PA \sin A = PB \sin B$, and therefore,

$$PA \sin A = PB \sin B = PC \sin C$$

: Let us denote by $D : E : F$ the foot of the perpendicular line drawn from P to the line segment BC, CA and AB, respectively. Since the points E, F lie on a circle having PA as its diameter, we have by the law of sines $EF = PA \sin A$: Similarly, we have $FD = PB \sin B$ and $DE = PC \sin C$.
Consequently, we conclude that DEF is an equilateral triangle. Furthermore, we have $\angle CPE = \angle CDE$, since the quadrilateral $CDPE$ is cyclic. Similarly, we have $\angle FPB = \angle FDB$. Putting these together, we get;

$$\begin{aligned} \angle BPC &= 360^0 - (\angle CPE + \angle FPB + \angle EPF) \\ &= 360^0 - \{(\angle CDE + \angle FDB) + (180^0 - \angle FAE)\} \\ &= 360^0 - (120^0 + 150^0) \\ &= 90^0. \end{aligned}$$

which proves the assertion of the problem.

Alternate Solution:
Let O be the midpoint of the line segment B_1B_2. Then the points B and P lie on the circle with center at O and going through the point B_1. From

$$\angle OBC = \angle OBB_1 - \angle CBB_1 = \angle OB_1B - \angle B_1BA = \angle BAC$$

it follows that the triangles OCD and OBA are similar, and therefore we have that

$$OC \cdot OA = OB^2 = OP^2$$

Thus we conclude that the triangles OCP and OPA are similar, and therefore, we have $\angle OPC = \angle PAC$: Using this fact, we obtain

$$\begin{aligned} \angle PBC - \angle PBA &= (\angle B_1BC + \angle PBB_1) - (\angle ABB_1 - \angle PBB_1) \\ &= 2\angle PBB_1 = \angle POB_1 = \angle PCA - \angle OPC \\ &= \angle PCA - \angle PAC \end{aligned}$$

from which we conclude that $\angle PAC+\angle PBC = \angle PBA+\angle PCA$ *Similarly, we get* $\angle PAB + \angle PCB = \angle PBA + \angle PCA$. *Putting these facts together and taking into account the fact that*

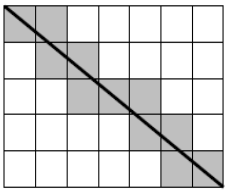
$$(\angle PAC + \angle PBC) + (\angle PAB + \angle PCB) + (\angle PBA + \angle PCA) = 180^0$$

we conclude that $\angle PBA + \angle PCA = 60^0$ *and finally that*

$$\angle BPC = (\angle PBA+\angle PAB)+(\angle PCA+\angle PAC) = \angle BAC+(\angle PBA+\angle PCA) = 90^0$$

proving the assertion of the problem. ■

Qn. 13. A diagonal of this 5×7 rectangle passes through 11 squares. (We say that a line passes through a square if the square and the line have at least two common points.) How many



square will the diagonal of a 2016×2018 rectangle pass through?

Solution : *First, we show that if we consider a rectangle with integer sides k and l where k and l are relatively prime positive integers with $k < l$, then any of its diagonals passes through $k + l - 1$ squares. We can assume that the bottom left corner of the rectangle is the origin of the coordinate system $(0, 0)$ and that the the remaining three vertices of the rectangle are $(l, 0)$, (l, k) and $(0, k)$. Also, let us consider the diagonal with endpoints $(0, k)$ and $(l, 0)$. Since the absolute value of the slope of the diagonal is less than 1, for each $t = 0, 1, \ldots, l - 1$ the diagonal passes either through one or two of the squares which are in the strip between the vertical lines $x = t$ and $x = t + 1$. It passes through one square if the diagonal does not intersect any of the horizontal lines $y = 1, y = 2, \ldots y = k - 1$ and it passes through two squares when the diagonal does intersect one of these lines.*

Thus, the number of squares the diagonal passes through equals the number of vertical strips (which is k) + the number of intersections of the diagonal and the horizontal lines $y = 1$, $y = 2, \ldots$, $y = k - 1$ (which is $k - 1$). So, indeed when k and l are relatively prime a diagonal passes through $k + l - 1$ squares.

Next, consider a 2016×2018 rectangle. We can assume that it has vertices $(0, 0)$, $(2018, 0)$, $(2018, 2016)$, $(0, 2016)$ and that the diagonal has endpoints $(0, 2016)$ and $(2018, 0)$. The center of the rectangle is $(1009, 1008)$. Since 1009 and 1008 are relatively prime, the diagonal passes through $1008 + 1009 - 1 = 2016$ squares of the rectangle with vertices $(0, 1008)$, $(1009, 1008)$, $(1009, 2016)$, $(0, 2016)$; it passes through $1008 + 1009 - 1 = 2016$ squares of the rectangle with vertices $(1009, 0)$, $(2018, 0)$, $(2018, 1008)$, $(1009, 1008)$; and it passes through no other squares. Therefore, passes through 4032 rectangles. ■

Qn. 14. (a) Suppose f is a function such that for every real number x, $f(x) + f(1-x) = 10$ and $f(1+x) = 4 + f(x)$. What is the value of $f(100) + f(-100)$?

Solution : *Replace x by $-x$ in the second property, and get $f(1-x) = 4 + f(-x)$. Substitute this in the first property and get $f(x) + 4 + f(-x) = 10$ for all x.*
$\Rightarrow f(100) + f(-100) = 6$ ■

(b) Find all positive real numbers x and y that satisfy the following system of equations:

$$\begin{aligned} x^y &= y^{x-y} \\ x^x &= y^{12y} \end{aligned}$$

Solution :

$$ylnx = (x-y)lny \qquad lnx = \frac{(x-y)}{y}lny$$

$$xlnx = 12ylny \qquad \frac{x(x-y)}{y}lny = 12ylny$$

$$\Leftrightarrow$$

$$lnx = \frac{x-y}{y}lny$$

$$\left(\frac{x(x-y)}{y} - 12y\right)lny = 0$$

From the last equation either $lny = 0$ or $\dfrac{x^2 - xy - 12y^2}{y} = 0$

If $lny = 0$, then $lnx = 0$. Therefore, $x = 1$, $y = 1$ is a solution.

If $\dfrac{x^2 - xy - 12y^2}{y} = 0$, then $\dfrac{x^2}{y^2} - \dfrac{x}{y} - 12 = 0$ and $\left(\dfrac{x}{y} - 4\right)\left(\dfrac{x}{y} + 3\right) = 0$

Since x and y are positive numbers, $x = 4y$ and

$ln(4y) = 3ln(y)$. Therefore, $4y = y^3$, $y = 2$, and $x = 8$.

Thus, the system has two solutions

x=1	*and*	*x=8*
y=1		*y=2*

■

Qn. 15. You and your partner went to a dinner party in which there were four other couples. After the dinner was over, you asked everyone except yourself: "How many people did you shake hands with tonight?" To your surprise, no two people gave the same number, so that someone did not shake any hands, someone else shook only one person's hand, a third person only shook two people's hands, and so on. Assume nobody shook hands with their own partner or with themselves. How many people did your partner shake hands with that evening at the party?

Solution : *The maximum possible number of people whose hands anyone shook is 8, and the only way* 9 *people can give different answers if all integers from* 0 *through* 8 *are included. Let A be the one who shook the hands of* 8 *people. Since everyone shook hands with A, except for A's partner and A him- or herself, then A's partner has to be the one who shook* 0 *people's hands*
Similar reasoning shows that the partner of the one who shook 7 *people's hands, shook only* 1 *person's hand (specifically, A's hand). The partner of the one who shook* 6 *(resp.* 5*) people's hands, shook* 2 *(resp.* 3*) people's hands.*
All these people are distinct from yourself, and also from your partner. This leaves 4 *for the answer given by your partner.*
Incidentally, with this information, it is possible to deduce that you also shook hands with 4 *people- the ones who shook* $8, 7, 6$ *and* 5 *people's hands.* ■